

An empirical example: *Rhinoclemmys* data

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I conducted a phylogenetic diversity (PD) analysis for the *Rhinoclemmys* genus to evaluate the effect of branch length on the chosen conservation area(s). The topology used corresponds to a total evidence analysis from Romero-Alarcon (2020), and the distribution was modified from Le and McCord (2008).

Read the data: the distribution, and the tree

```
## For reproductibility purposes

set.seed(121)

options(warn=-1)
library(ggplot2)
suppressMessages(library(ggtree))

suppressMessages(library(blepd))

suppressMessages(library(phytools))

#~ library(ggtree)
##
## If you want to install ggtree.
##
### https://bioconductor.org/packages/release/bioc/html/ggtree.html
##
####
# if (!require("BiocManager", quietly = TRUE))
#   install.packages("BiocManager")
# BiocManager::install("ggtree")

## Version

cat("Analyses made with blepd version:", unlist(packageVersion("blepd"))))

## Analyses made with blepd version: 0 3 2 2024 7 30
## Create an object to place the distribution and the tree

RhinoclemmysData <- list()

## Read data
```

```

## distribution is a labeled csv file, areas by terminals

#setwd("./csv/")

## getwd()

csvFile <- list.files(pattern="csv")

## csvFile

### The functions use a matrix object for the distributions

RhinoclemmysData$distribution <- as.matrix(read.table(csvFile,
  stringsAsFactors=FALSE,
  header=TRUE,
  row.names=1,
  sep=",")
)

print(t(RhinoclemmysData$distribution))

##           Al An-As Cho Nsa Nuh Ph Pl
## R_annulata    1     0  1  0  0  1  0
## R_areolata    1     0  0  0  0  0  0
## R_diademata    0     0  0  1  0  0  0
## R_funerea      1     0  0  0  1  0  0
## R_melanosterna 0     0  1  0  0  1  0
## R_nasuta       0     0  1  0  0  0  0
## R_pulcherrima  0     0  0  0  1  0  1
## R_punctularia  0     1  0  0  0  0  0
## R_rubida       0     0  0  0  0  0  1

## Tree(s) in nexus or newick format

##setwd("../tree/")

treeFiles <- list.files(pattern=".tre")

##treeFiles

RhinoclemmysData$tree <- read.tree(treeFiles)

## name of tree(s)

NameTreeFile <- gsub(".tre","",treeFiles)

## Plotting:

## The tree

```

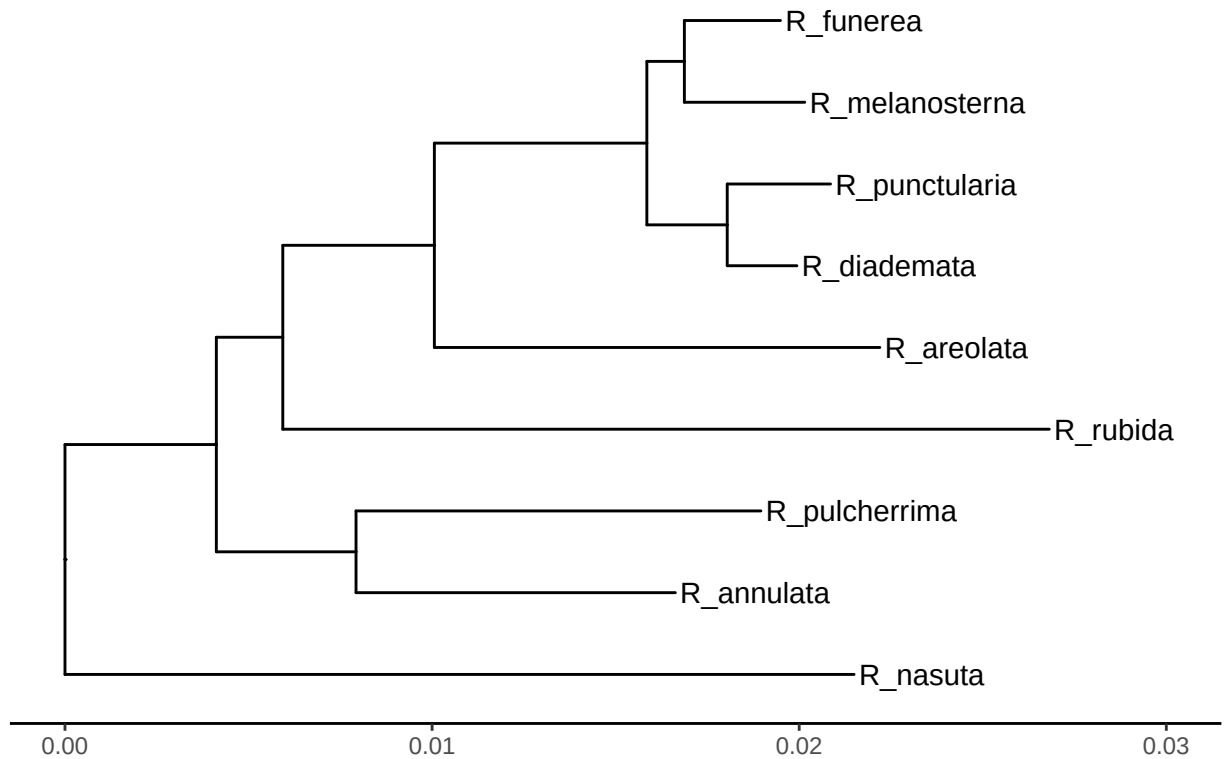
```
## using ggtree

# RhinoclemmysData$tree <- reorder(RhinoclemmysData$tree, order = "cladewise")

plotTree <- ggtree(RhinoclemmysData$tree, ladderize=TRUE,
                  color="black", size=0.51, linetype="solid") +
  geom_tiplab(size=4, color="black") +
  xlim(0,0.030) +
  theme_tree2() +
  ggtitle(treeFiles[1])

print(plotTree)
```

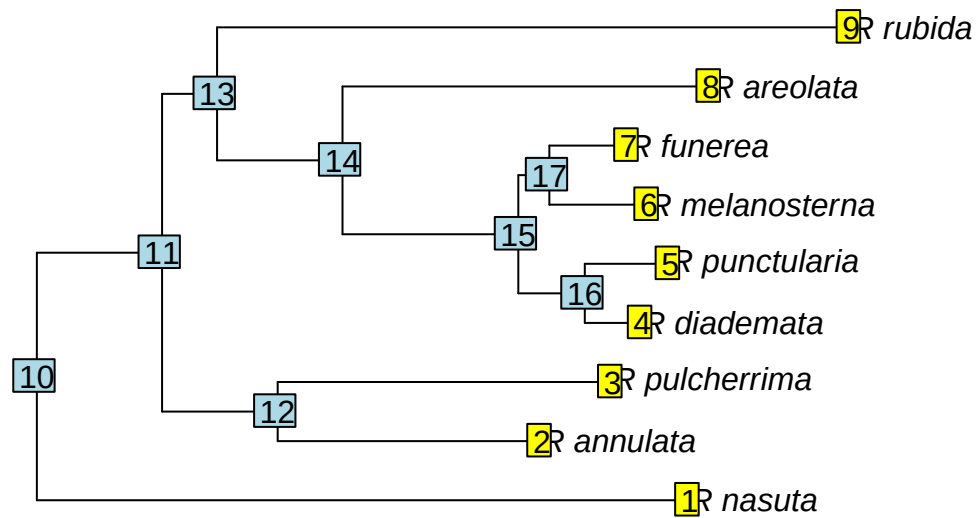
Rhinoclemmys_TotalEvidence_ML.tre



```
#~ Alternatively, we can plot the tree using APE

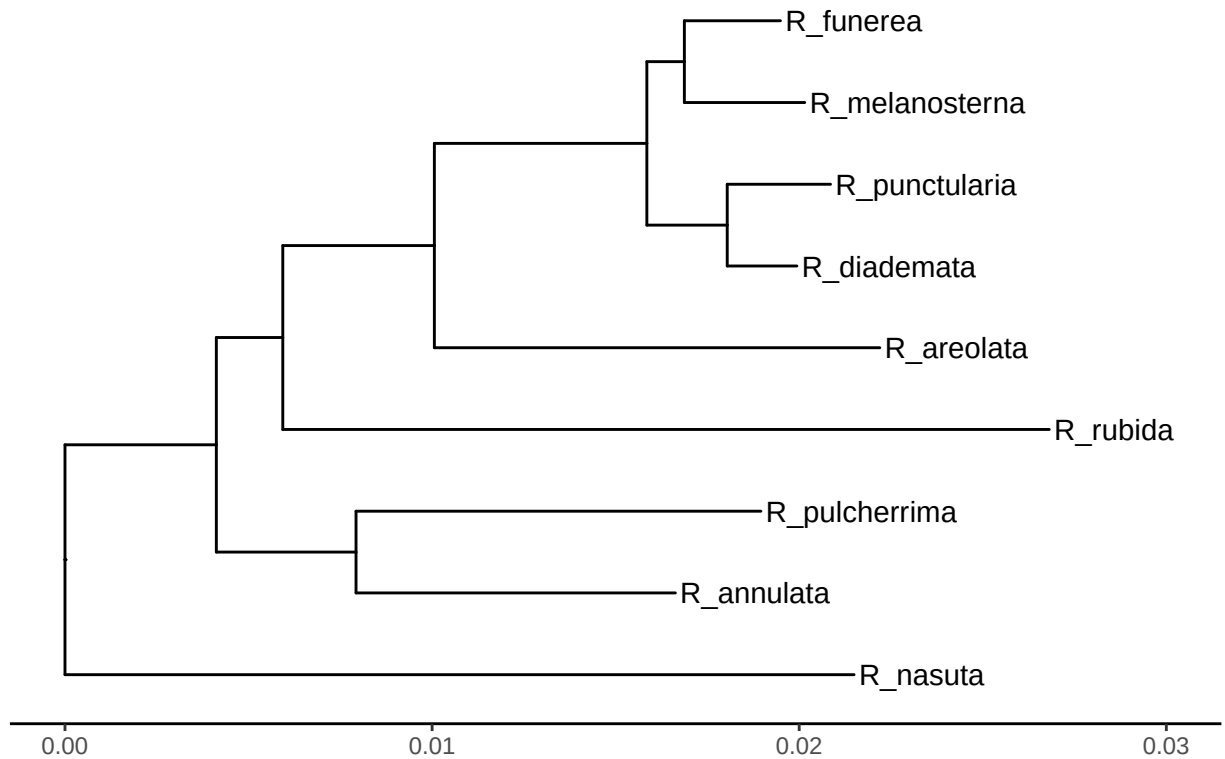
plot.phylo(RhinoclemmysData$tree)

# The number of the nodes / tips
#
nodelabels()
####
tiplabels()
```



```
plot(plotTree)
```

Rhinoclemmys_TotalEvidence_ML.tre



```
print(t(RhinoclemmysData$distribution))
```

```
##           Al An-As  Cho  Nsa  Nuh  Ph  Pl
## R_annulata    1     0   1   0   0   1   0
## R_areolata    1     0   0   0   0   0   0
## R_diademata    0     0   0   1   0   0   0
## R_funerea      1     0   0   0   1   0   0
## R_melanosterna  0     0   1   0   0   1   0
## R_nasuta       0     0   1   0   0   0   0
## R_pulcherrima  0     0   0   0   1   0   1
## R_punctularia  0     1   0   0   0   0   0
## R_rubida       0     0   0   0   0   0   1
```

Now we plot the branch lengths

```
library(ggplot2)

##par(mfrow=c(2,1))

datos <- RhinoclemmysData$tree$edge.length
data <- as.data.frame(datos)

a <- ggplot2::ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(title = treeFiles, x = "Branch length: internals and terminals", y = "Frequency") +
```

```

      xlim(c(0.001, 0.025)) +
      ylim(c(0, 5))

terminals <- RhinoclemmysData$tree$edge[,2] < 1 + length(RhinoclemmysData$tree$tip.label)

datos <- RhinoclemmysData$tree$edge.length[terminals]
data <- as.data.frame(datos)

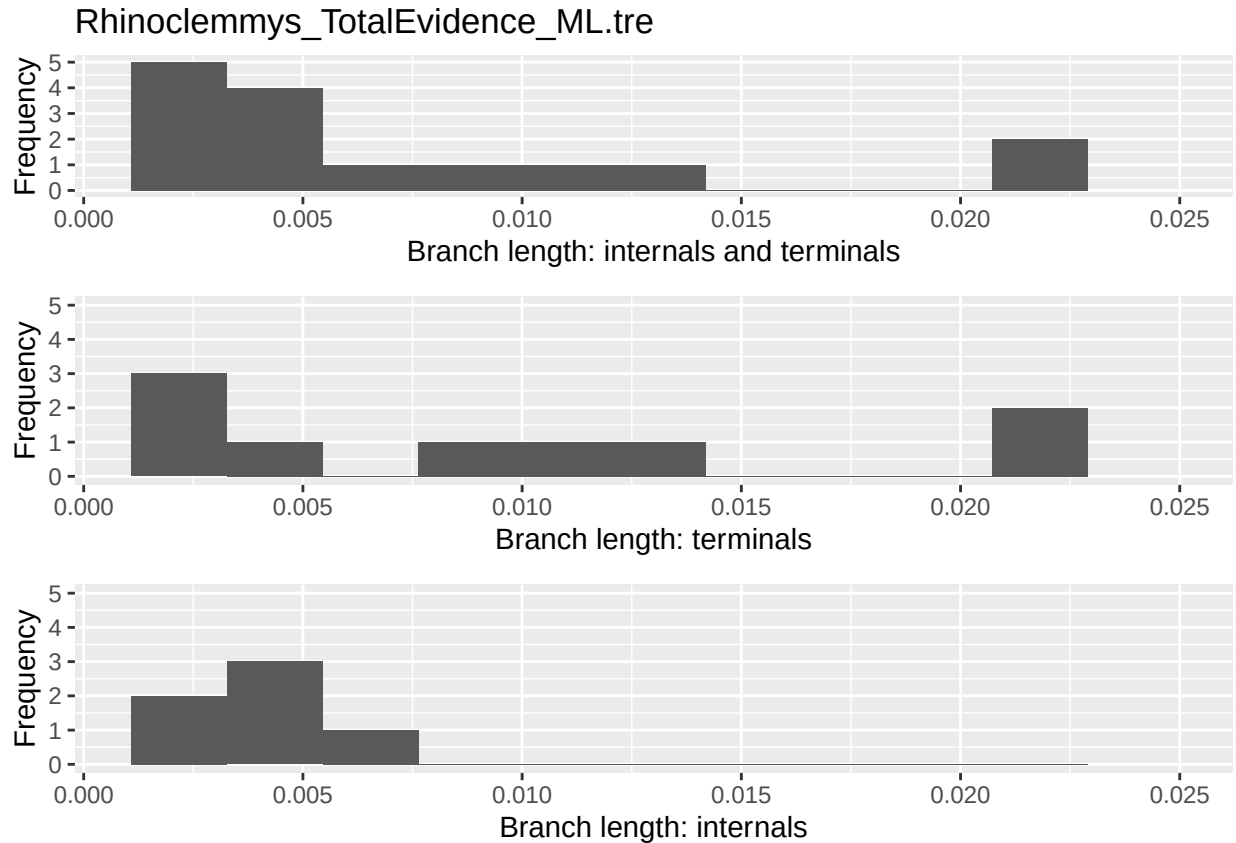
b <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: terminals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))

datos <- RhinoclemmysData$tree$edge.length[!terminals]
data <- as.data.frame(datos)

c <- ggplot(data, aes(x=datos)) +
  geom_histogram(bins = 12) +
  labs(x = "Branch length: internals", y = "Frequency") +
  xlim(c(0.001, 0.025)) +
  ylim(c(0, 5))

cowplot::plot_grid(a, b, c, nrow=3)

```



```
#
# cat("Terminals, with BL larger than 0.02 = ",
# findTerminalgivenLength(RhinoclemmysData$tree,0.02))
#
```

The branch length histograms and the tree plot showed that the internal length branches were similar among each other, and different to terminals'; there were two longer branches (larger than 0.02), *R. nasuta* (inhabiting Cho) and *R. rubida* (Pl), whereas Al and Cho were the richest.

PD values

First, I calculated the PD value for the areas, given this tree.

```
RhinoclemmysData$tablePD <- PDindex(RhinoclemmysData$tree,
                                     distribution=RhinoclemmysData$distribution,
                                     root=TRUE,
                                     percentual = TRUE)

RhinoclemmysData$matrixPD <- as.data.frame(
  matrix(
    unlist(RhinoclemmysData$tablePD),
    nrow= length(treeFiles),
    byrow=TRUE))

colnames(RhinoclemmysData$matrixPD) <- row.names(RhinoclemmysData$distribution)
```

```
row.names(RhinoclemmysData$matrixPD) <- "PD value"
```

```
new <- (t(RhinoclemmysData$matrixPD))
```

```
values <- new[order(new[,1])]
```

```
names <- rownames(new)[order(new[,1])]
```

```
print(as.data.frame(paste(names,values)))
```

```
## paste(names, values)
```

```
## 1          Nsa 8.05
```

```
## 2        An-As 8.42
```

```
## 3          Ph 13.18
```

```
## 4          Nuh 13.86
```

```
## 5          Pl 16.81
```

```
## 6          Al 17.82
```

```
## 7          Cho 21.86
```

The highest PD was for area Cho, followed by Al, the two richest areas. However, the difference in PD values was not determined by species richness alone, but also by the branch lengths of the species within each region.

#Effect of branch lengths in the PD

I tested the effect of branch length on PD values by swapping terminal and/or internal branch lengths using three different models.

```
for( modelo in c("simpleswap","allswap","uniform") ){

  for( rama in c("terminals","internals","all") ){

    val <- swapBL(RhinoclemmysData$tree,
                  RhinoclemmysData$distribution,
                  model = modelo,
                  branch = rama
                )

    cat( "\n\t Tree=",treeFiles,"\n\t Model=",modelo,
          "\n\t Branchs swapped=",rama,"\n" )

    printSwapBL(val)

    cat( "\n\n" )

  }

}
```

```
## model to test simpleswap reps 100
```

```
##
```

```
## Tree= Rhinoclemmys_TotalEvidence_ML.tre
```

```
## Model= simpleswap
```

```
## Branchs swapped= terminals
```

```
## Initial      Branch      Model Al Cho Pl
```

```
## 1      Cho terminals simpleswap 13 85 2
```



```

##
##
## model to test simpleswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML.tre
##   Model= simpleswap
##   Branchs swapped= internals
##   Initial   Branch       Model Cho
## 1      Cho internals simpleswap 100
##
##
## model to test simpleswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML.tre
##   Model= simpleswap
##   Branchs swapped= all
##   Initial Branch       Model Al Cho Pl
## 1      Cho    all simpleswap 8 91 1
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML.tre
##   Model= allswap
##   Branchs swapped= terminals
##   Initial   Branch   Model Al Cho Nuh Pl
## 1      Cho terminals allswap 35 53 10 2
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML.tre
##   Model= allswap
##   Branchs swapped= internals
##   Initial   Branch   Model Cho
## 1      Cho internals allswap 100
##
##
## model to test allswap reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML.tre
##   Model= allswap
##   Branchs swapped= all
##   Initial Branch   Model Al An-As Cho Nsa Nuh Pl
## 1      Cho    all allswap 37    4 33  4 19 3
##
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML.tre
##   Model= uniform
##   Branchs swapped= terminals
##   Initial   Branch   Model Al Cho Nuh
## 1      Cho terminals uniform 46 50 4

```

```
##
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML.tre
##   Model= uniform
##   Branchs swapped= internals
##   Initial   Branch   Model Cho
## 1      Cho internals uniform 100
##
##
## model to test uniform reps 100
##
##   Tree= Rhinoclemmys_TotalEvidence_ML.tre
##   Model= uniform
##   Branchs swapped= all
##   Initial Branch   Model Al An-As Cho Nsa Nuh
## 1      Cho    all uniform 38      1 53  1  7
```

I determined that terminal branch lengths were crucial in influencing the chosen area. When I swapped all terminal branch lengths (model = “allswap”) or replaced them with a uniform distribution (model = “uniform”), the optimal area shifted from Cho to Al. Recognizing the uneven distribution of terminal branch lengths, I suspected that the longest branches within the Cho and/or Al areas were driving the results.

I first investigated the impact of the number of replicates on the results.

```
for(repetir in 1:5){

    val <- swapBL( RhinoclemmysData$tree,
                   RhinoclemmysData$distribution,
                   model = "allswap",
                   branch = "all",
                   nTimes = 10**repetir
                 )

    printSwapBL(val)

}
```

```
## model to test allswap reps 10
##   Initial Branch   Model Al An-As Cho Nsa Nuh
## 1      Cho    all allswap 40      10 30  10 10
## model to test allswap reps 100
##   Initial Branch   Model Al An-As Cho Nsa Nuh Pl
## 1      Cho    all allswap 26      3 50  4 13  4
## model to test allswap reps 1000
##   Initial Branch   Model Al An-As  Cho Nsa  Nuh  Pl
## 1      Cho    all allswap 37    3.9 39.2 4.6 13.4 1.9
## model to test allswap reps 10000
##   Initial Branch   Model    Al An-As    Cho Nsa    Nuh  Pl
## 1      Cho    all allswap 36.55  4.33 38.87 4.43 13.96 1.86
## model to test allswap reps 1e+05
##   Initial Branch   Model    Al AlNsa An-As    Cho Nsa    Nuh  Pl
## 1      Cho    all allswap 36.21      0  4.42 39.86 4.48 13.43 1.6
```

I found that results stabilized after approximately 1000 replicates, with the most substantial differences

arising from using 10 or 100 replicates. As a general guideline, at least 100 replicates are necessary, but 1000 are preferable.

Next, I investigated whether the observed differences in results could be attributed to the length of the longest branches.

```
## we can use a single command (*evalBranch*)

AllTerminals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="terminals",
                           approach="all",
                           printNames=TRUE)
```

```
printEvalBranch(AllTerminals)
```

```
##           node initialArea FinalArea Aproach   %Delta
## 1      [R_nasuta]|1          Cho      A1    lower -45.859
## 12 [R_pulcherrima]|3          Cho      P1    upper 115.809
## 13  [R_diademata]|4          Cho      Nsa    upper 1710.405
## 14 [R_punctularia]|5          Cho     An-As    upper 1103.961
## 16   [R_funerea]|7          Cho      A1    upper  325.502
## 17  [R_areolata]|8          Cho      A1    upper   86.219
## 18   [R_rubida]|9          Cho      P1    upper   59.024
```

```
AllInternals <- evalBranch(tree=RhinoclemmysData$tree,
                           distribution=RhinoclemmysData$distribution,
                           branchToEval="internals",
                           approach="all",
                           printNames=TRUE)
```

```
printEvalBranch(AllInternals)
```

```
##           node initialArea FinalArea Aproach
## 13 [node number:16:R_diademata/R_punctularia]|16          Cho     An-As    upper
##           %Delta
## 13 1694.723
```

I found that branch length significantly influenced the selected area. While modifying internal branches had minimal effects, altering terminal branch lengths substantially impacted area selection. Specifically, a 46% decrease in *R. nasuta*'s or an 86% increase in *R. areolata*'s terminal branch length shifted the optimal area from Cho to A1. Similarly, a 59% increase in *R. rubida*'s or a 116% increase in *R. pulcherrima*'s terminal branch length resulted in a shift from Cho to P1.

As an additional exercise, I investigated the impact of ultrametricity. I hypothesized that if the tree were ultrametric, the terminal sister pairs would behave similarly, with differences primarily attributable to the underlying distribution.

```
## now let's see the impact of ultra-metricity

AllTerminalsU <- evalBranch(tree=force.ultrametric(RhinoclemmysData$tree),
                            distribution=RhinoclemmysData$distribution,
                            branchToEval="terminals",
                            approach="all",
                            printNames=TRUE)
```

```
## *****
## *                               Note:                               *
```

```
## * force.ultrametric does not include a formal method to *
## * ultrametricize a tree & should only be used to coerce *
## * a phylogeny that fails is.ultrametric due to rounding -- *
## * not as a substitute for formal rate-smoothing methods. *
## *****
```

```
cat("Ultrametric tree\n")
```

```
## Ultrametric tree
```

```
printEvalBranch(AllTerminalsU)
```

```
##           node initialArea FinalArea Approach %Delta
## 9      [R_nasuta]|1          Cho        A1    lower -48.953
## 11     [R_funerea]|7          Cho        A1    upper 336.352
## 12    [R_diademata]|4          Cho        Nsa    upper 1890.607
## 13 [R_punctularia]|5          Cho       An-As    upper 1890.607
## 14    [R_areolata]|8          Cho        A1    upper  92.029
## 15      [R_rubida]|9          Cho        P1    upper 134.869
## 17 [R_pulcherrima]|3          Cho       NuhP1    upper 209.954
```

```
printEvalBranch(AllInternals)
```

```
##           node initialArea FinalArea Approach %Delta
## 1      [R_nasuta]|1          Cho        A1    lower -45.859
## 12 [R_pulcherrima]|3          Cho        P1    upper 115.809
## 13    [R_diademata]|4          Cho        Nsa    upper 1710.405
## 14 [R_punctularia]|5          Cho       An-As    upper 1103.961
## 16     [R_funerea]|7          Cho        A1    upper 325.502
## 17    [R_areolata]|8          Cho        A1    upper  86.219
## 18      [R_rubida]|9          Cho        P1    upper  59.024
```

```
AllInternalsU <- evalBranch(tree=force.ultrametric(RhinoclemmysData$tree),
                             distribution=RhinoclemmysData$distribution,
                             branchToEval="internals",
                             approach="all",
                             redondeo=3,
                             printNames=TRUE)
```

```
## *****
## * Note: *
## * force.ultrametric does not include a formal method to *
## * ultrametricize a tree & should only be used to coerce *
## * a phylogeny that fails is.ultrametric due to rounding -- *
## * not as a substitute for formal rate-smoothing methods. *
## *****
```

```
printEvalBranch(AllInternalsU)
```

```
##           node initialArea FinalArea Approach
## 8 [node number:16:R_diademata/R_punctularia]|16          Cho An-AsNsa upper
## %Delta
## 8 1860.13
```

```
printEvalBranch(AllInternals)
```

```
##           node initialArea FinalArea Approach
## 13 [node number:16:R_diademata/R_punctularia]|16          Cho An-As upper
```

%Delta
13 1694.723

These findings suggest that decisions to conserve solely area Cho may be misguided. Areas Al, and potentially Pl, should also be considered as essential components of the conservation range.

Literature cited

- Le, Minh, and William P. McCord. 2008. "Phylogenetic relationships and biogeographical history of the genus *Rhinoclemmys* Fitzinger, 1835 and the monophyly of the turtle family Geoemydidae (Testudines: Testudinoidea)." *Zoological Journal of the Linnean Society* 153 (4): 751–67. <https://doi.org/10.1111/j.1096-3642.2008.00413.x>.
- Romero-Alarcon, L. Viviana. 2020. "A total-evidence phylogeny of the crown and stem-groups of turtles (Pantestudines: Testudinata)." Master's thesis, Colombia: Escuela de Biología, UIS.